

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<answer> A . Neonatal respiratory distress syndrome (NRDS) </answer>

<think> Reasoning process :

- 1 . **Birth at 32 nd gestational week** : Premature birth is a significant risk factor for neonatal respiratory distress syndrome (NRDS).
- 2 . **Symptoms** : The baby is experiencing tachypnea , subcostal and intercostal retractions , nasal flaring , and cyanosis . These signs are indicative of respiratory distress , which is a hallmark of NRDS .
- 3 . **Response to oxygen** : The cyanosis responded well to initial oxygen administration , suggesting that the baby 's primary issue is respiratory rather than cardiovascular .
- 4 . **Absence of other significant signs or symptoms** : There 's no mention of abnormal heart rate , breathing patterns , or other signs that would suggest a congenital heart anomaly or infection like pneumonia .
- 5 . **Presence of gestational diabetes** : While gestational diabetes can increase the risk of complications during pregnancy , the baby 's respiratory distress symptoms do not directly suggest an issue related to diabetes .
- 6 . **Nasogastric tube positioning** : This suggests that the baby has good respiratory function initially , as it wasn 't challenging to position .

Considering these factors , the most likely diagnosis is neonatal respiratory distress syndrome (NRDS), which is a common cause of respiratory distress in preterm infants . This condition typically arises due to the immaturity of the lungs ' surfactant production , leading to difficulty in gas exchange . The symptoms presented align well with NRDS , although a definitive diagnosis would require further evaluation , including tests such as chest X-rays and blood gas analysis to assess the severity of lung function and oxygenation .